WORKED SOLUTION

STEP III 1999, Question 10

Mechanics • Falling Chain Impulse

STEP III 1999, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s3-q10

The key idea

While the chain collapses, the part still hanging is in free fall: nothing pulls on it except gravity. So each link simply drops from wherever it started, and the only thing that varies from link to link is the height of that drop. The table supplies one impulse per link to stop it. Adding these up gives a sum of square roots, and for large n this is a Riemann sum for $\int_0^1 \sqrt{x}\,\mathrm{d}x$.

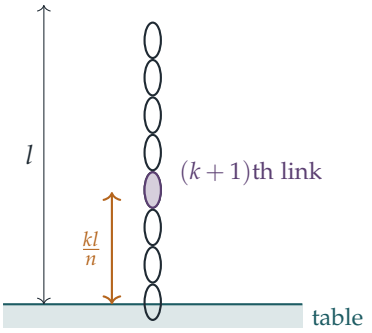
Solution

The model

Each link has mass $\frac{m}{n}$ and length $\frac{l}{n}$. The links are smooth, and the chain is perfectly flexible, so it can't transmit a push. We treat each link as a particle that is brought to rest instantly (the collision is inelastic) when it reaches the table.

? **Ask yourself** Why is the hanging part of the chain in free fall, when the links below it are being stopped by the table?

When the chain is released, every link has acceleration g if there is no tension, and a falling chain with zero tension is consistent: all links move together and stay in contact without stretching. The table stops a link by pushing it up, and a flexible chain can't pass that push on to the links above it. Since the links are smooth, there is no friction between them either. So no force other than gravity acts on any link that is still in the air, and each link falls freely until it lands.



The $(k+1)$th link

Below the $(k+1)$th link there are k links, with total length $\frac{kl}{n}$. As the bottom end just touches the table, the $(k+1)$th link falls a distance $\frac{kl}{n}$ before it reaches the table. Falling freely from rest, its speed on arrival is given by $v^2 = 2gs$:

$$v_k = \sqrt{\frac{2gkl}{n}}.$$

It is then brought to rest, so the magnitude of its change in momentum is

$$\frac{m}{n}\sqrt{\frac{2gkl}{n}} = \frac{m}{n}\sqrt{2gl}\,\sqrt{\frac{k}{n}},$$

directed upwards (the link loses downward momentum $\frac{m}{n}v_k$). For $k = 0$ this is zero: the bottom link is already touching the table.

The total impulse

By Newton's third law, the impulse the table sustains from the $(k+1)$th link is equal in size to that link's change in momentum, and points downwards. The links are numbered $k+1 = 1, 2, \ldots, n$, so the total impulse on the table from the collisions is

$$J = \sum_{k=0}^{n-1} \frac{m}{n}\sqrt{\frac{2gkl}{n}} = m\sqrt{2gl}\ \cdot\ \frac{1}{n}\sum_{k=0}^{n-1}\sqrt{\frac{k}{n}}.$$

? **Ask yourself** Which integral does $\frac{1}{n}\sum_{k=0}^{n-1}\sqrt{k/n}$ approximate?

It is the sum of the areas of n rectangles of width $\frac{1}{n}$ whose heights are the values of $\sqrt{x}$ at the left-hand ends $x = 0, \frac{1}{n}, \ldots, \frac{n-1}{n}$ of the strips of $[0,1]$. So for large n,

$$\frac{1}{n}\sum_{k=0}^{n-1}\sqrt{\frac{k}{n}} \approx \int_0^1 \sqrt{x}\,\mathrm{d}x = \left[\tfrac{2}{3}x^{3/2}\right]_0^1 = \frac{2}{3},$$

and therefore

$$J \approx \tfrac{2}{3}\,m\sqrt{2gl},$$

as required.

To see how good the approximation is: $\sqrt{x}$ is increasing, so on the strip $[\frac{k}{n}, \frac{k+1}{n}]$ the left-end height $\sqrt{k/n}$ lies below the curve, and on $[\frac{k-1}{n}, \frac{k}{n}]$ it lies above it. Summing over the strips gives

$$\int_0^{1-\frac{1}{n}} \sqrt{x}\,\mathrm{d}x \;\le\; \frac{1}{n}\sum_{k=0}^{n-1}\sqrt{\frac{k}{n}} \;\le\; \int_0^1 \sqrt{x}\,\mathrm{d}x, \quad \text{that is} \quad \tfrac{2}{3}\left(1-\tfrac{1}{n}\right)^{3/2} \le \frac{J}{m\sqrt{2gl}} \le \tfrac{2}{3}.$$

Both bounds tend to $\frac{2}{3}$, so the relative error is at most about $\frac{3}{2n}$.

Checks

- Units: $m\sqrt{gl}$ is mass $\times$ speed, which is a momentum, as an impulse must be.
- Numerically, with $m = 2$, $l = 3$, $g = 9.8$: $\frac{2}{3}m\sqrt{2gl} \approx 10.224$, while the exact sum is 9.363 for $n = 10$, 10.144 for $n = 100$ and 10.216 for $n = 1000$. The ratio approaches 1, and always lies inside the bounds above.
- The last link to land falls almost the whole length l, with speed close to $\sqrt{2gl}$. If all the mass arrived at that speed the impulse would be $m\sqrt{2gl}$. Most links fall less far, so it makes sense that the answer is a fraction of this.

Common slips

- Letting the $(k+1)$th link fall a distance $\frac{(k+1)l}{n}$. There are only k links beneath it.
- Including the steady weight of the chain already lying on the table. The question asks only for the impulse from the collisions.

Beyond STEP

Since all the links land in time $\sqrt{2l/g}$, the collisions alone give the table an average force of $\frac{2}{3}m\sqrt{2gl}/\sqrt{2l/g} = \frac{2}{3}mg$. At each moment the impact force is actually twice the weight of chain already on the table, so the reading on a scale under the table climbs to $3mg$ just as the last link lands. Problems like this, where mass is gained or lost continuously (falling chains, rockets, raindrops), are treated with the variable-mass form of Newton's second law. See Kleppner and Kolenkow, *An Introduction to Mechanics*, for a careful treatment.

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